STPM Physics · Semester 2

DEFINITION LIST

Electricity & Magnetism · Chapters 13–19. 188 definitions, laws and statements; key words in bold. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 13 — ELECTROSTATICS

13.1 Coulomb's Law

1. **Coulomb's law** states that the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.
2. The force acts along the line joining the two charges; it is repulsive for like charges and attractive for unlike charges. The two forces form a Newton's third law pair (equal in magnitude, opposite in direction).
3. **Principle of superposition**: the net force on a charge is the vector sum of the forces exerted on it by each of the other charges separately.
4. **Relative permittivity** ε_r of a medium is the ratio of the force between two charges in a vacuum to the force between the same charges, at the same separation, in the medium ($\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$).
5. **Charge is quantised**: Q = ne, where n is an integer.

13.2 Electric Field

6. An **electric field** is a region in which an electric force acts on a charged body.
7. **Electric field strength** at a point is the force per unit positive charge placed at that point ($E = \frac{F}{q}$); it is a vector quantity whose direction is that of the force on a positive test charge.
8. **Electric field line**: a line whose tangent at any point gives the direction of the electric field at that point; the density of the lines is proportional to the field strength.
9. Field lines begin on positive charges (or at infinity) and end on negative charges (or at infinity); they never cross, because the field has only one direction at a point.
10. The field of a point charge is radial: outward for a positive charge and inward for a negative charge.

11 A **uniform field** exists between two oppositely charged parallel plates: the field lines are parallel and equally spaced, and E has the same value at every point (apart from the edges).

12 **Superposition**: the resultant field at a point is the vector sum of the fields due to the individual charges.

13.3 Gauss's Law

13 **Electric flux** through a surface is the product of the electric field strength normal to the surface and the area of the surface, $\Phi = EA\cos\theta$, where θ is the angle between E and the normal to the surface.

14 **Gauss's law** states that the total electric flux through any closed surface is equal to the net charge enclosed by the surface divided by ε_0.

15 The flux depends only on the net enclosed charge, not on the shape or size of the closed surface, and not on the position of the charge inside the surface.

16 A charge outside the closed surface contributes zero net flux, because every field line that enters the surface also leaves it.

17 In electrostatic equilibrium the field inside the material of a conductor is zero, so the excess charge lies on the surface of the conductor.

18 **Surface charge density** σ is the charge per unit area of a surface, $\sigma = \frac{Q}{A}$.

19 Outside a charged sphere the field is the same as if all the charge were concentrated at the centre.

13.4 Electric Potential

20 **Electric potential** at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point ($V = \frac{W}{q}$).

21 **Potential difference** between two points is the work done per unit positive charge in moving a test charge from one point to the other.

22 **Electric potential energy** of a charge at a point is the work done in bringing the charge from infinity to that point ($U = qV$).

23 An **equipotential surface** is a surface on which all points are at the same potential; no work is done in moving a charge along it, and it is perpendicular to the electric field lines.

24 The electric field is the negative potential gradient ($E = -\frac{dV}{dr}$): the field points in the direction of decreasing potential, and its magnitude is large where equipotentials are close together.

25 Potential is a scalar: the potential due to several charges is the algebraic sum of the individual potentials.

26 One **electron-volt** (eV) is the energy gained by an electron when it is accelerated through a potential difference of 1 V; $1\text{ eV} = 1.60 \times 10^{-19}$ J.

CHAPTER 14 — CAPACITORS

14.1 Capacitance

1 **Capacitance** of a capacitor is the ratio of the magnitude of the charge on either plate to the potential difference between the plates ($C = \frac{Q}{V}$).

2 **One farad** is the capacitance of a capacitor that stores a charge of one coulomb when the potential difference across it is one volt ($1\ F = 1\ C\ V^{-1}$).

3 The capacitance depends only on the geometry of the capacitor (area, separation, shape) and the dielectric between the plates; it does not depend on the charge stored or the potential difference applied.

4 When a capacitor is connected to a battery the potential difference V is fixed; when it is isolated (disconnected) the charge Q remains constant.

14.2 Parallel Plate Capacitors

5 For a parallel plate capacitor the capacitance is directly proportional to the area of the plates and inversely proportional to their separation.

6 The electric field between the plates is uniform, $E = \frac{V}{d} = \frac{\sigma}{\varepsilon_0}$, directed from the positive plate to the negative plate; the edge effects are neglected when d is much smaller than the plate dimensions.

7 If the battery stays connected V is constant, so $Q = CV$ changes when d or A changes; if the capacitor is isolated Q is constant.

8 A slab of dielectric that partly fills the gap in layers (parallel to the plates) forms capacitors in series; a slab covering part of the area forms capacitors in parallel.

14.3 Dielectrics

9 A **dielectric** is an insulating material which becomes polarised in an electric field.

10 **Relative permittivity (dielectric constant)** ε_r is the ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance when there is a vacuum between them ($\varepsilon_r = \frac{C}{C_0} = \frac{\varepsilon}{\varepsilon_0}$).

11 **Polarisation:** in the external field the molecules of the dielectric become aligned (polar molecules) or stretched (non-polar molecules), so that opposite induced charges appear on the faces of the dielectric.

12 The induced charges produce an electric field opposite in direction to the applied field, so the net field in the dielectric is reduced ($E = \frac{E_0}{\varepsilon_r}$), the potential difference between the plates falls for the same charge and so the capacitance increases.

13 **Dielectric breakdown** occurs when the field exceeds the dielectric strength of the material and the insulator begins to conduct.

14.4 Capacitors in Series and in Parallel

14 In a **parallel** combination all capacitors have the same potential difference across them, and the total charge is the sum of the charges on the capacitors.

15 In a **series** combination every capacitor carries the same charge, and the total potential difference is the sum of the potential differences across the capacitors.

16 The equivalent capacitance in parallel is the sum of the capacitances; in series the reciprocal of the equivalent capacitance is the sum of the reciprocals.

17 In series the potential difference across a capacitor is inversely proportional to its capacitance, so the smaller capacitor has the larger potential difference.

14.5 Energy Stored in a Charged Capacitor

18 The **energy stored in a capacitor** is the work done in charging it, equal to the area under the graph of potential difference against charge.

19 **Energy density** of the electric field is the energy stored per unit volume of the field between the plates.

20 When a capacitor is charged from a battery through a resistor, the battery supplies energy QV, of which ½QV is stored in the capacitor and ½QV is dissipated as heat in the resistor, independent of R.

21 When a charged capacitor is connected to an uncharged capacitor the charge is conserved but the total energy decreases; the energy lost is dissipated as heat (and radiation).

14.6 Charging and Discharging of a Capacitor

22 The **time constant** $\tau = RC$ is the time taken for the charge on a discharging capacitor to fall to $\frac{1}{e}$ (about 37%) of its initial value; for charging it is the time for the charge to reach 63% $(1 - \frac{1}{e})$ of its final value.

23 During discharge the charge, potential difference and current decay exponentially with time; during charging the current decays exponentially while the charge and potential difference rise exponentially towards their final values.

24 At the start of charging the capacitor has no potential difference, so the initial current is $I_0 = \frac{V_0}{R}$; when the capacitor is fully charged the current is zero and the potential difference across it equals the EMF.

25 The time constant is increased by increasing R or C; the final charge depends on C and V_0 only.

▶ CHAPTER 15 — ELECTRIC CURRENT

15.1 Conduction of Electricity

1 **Electric current** is the rate of flow of charge through a cross-section of a conductor.

2 **Conduction in a metal** is due to the free (delocalised) electrons, which have large random thermal velocities in all directions; an applied p.d. sets up an electric field that superimposes a small net drift velocity opposite to the field.

3 **Conventional current** is taken in the direction of flow of positive charge, which is opposite to the direction of drift of the electrons.

4 **Explain why the resistance of a metal increases with temperature:** the lattice ions vibrate with greater amplitude, so the free electrons collide with them more frequently; the mean free time between collisions decreases while the free electron density is almost constant.

5 **Explain why the conductivity of a semiconductor increases with temperature:** more electrons are released from the covalent bonds, producing more electron-hole pairs; the increase in the number of charge carriers outweighs the increased lattice scattering.

15.2 Drift Velocity

6 **Drift velocity** is the **mean** velocity of the free electrons along the conductor due to the applied electric field, superimposed on their random thermal motion.

7 **Why a lamp lights almost instantaneously:** the electric field is established along the whole circuit at a speed close to that of light, so the free electrons throughout the wire, including those in the filament, begin to drift at once.

8 **In a steady current** the drift velocity is the same at every point of a uniform wire; in a wire of varying cross-section it is greater where the area is smaller.

9 **In a semiconductor** both electrons and holes contribute to the current; they drift in opposite directions but their contributions to the conventional current add.

15.3 Current Density

10 **Current density** is the current per unit cross-sectional area perpendicular to the direction of the current at that point, $J = \frac{I}{A}$.

11 **Current density** is a vector quantity with the direction of the conventional current.

12 **In a steady current** the current I is the same at every cross-section of a conductor, but the current density is greater where the cross-sectional area is smaller.

13 **Ohm's law (microscopic form):** the current density in a conductor is directly proportional to the electric field strength, $J = \sigma E$, at constant temperature.

14 **Analogy:** electric current corresponds to the rate of heat flow, and potential difference to temperature difference; conductivity corresponds to thermal conductivity.

15.4 Electric Conductivity and Resistivity

15 **Resistivity** of a material is the resistance of a wire of that material of unit length and unit cross-sectional area, $\rho = \frac{RA}{L}$ (unit: Ω m).

16 **Conductivity** is the reciprocal of resistivity, $\sigma = \frac{1}{\rho}$ (unit: $\Omega^{-1}\,m^{-1}$ or $S\,m^{-1}$); both are properties of the material, not of its size or shape.

17 **Conductance** is the reciprocal of resistance, $G = \frac{1}{R}$ (unit: siemens, $S = \Omega^{-1}$).

18 **Ohm's law**: the steady current in a metallic conductor is directly proportional to the potential difference across it, provided the physical conditions (such as temperature) remain constant.

19 A **superconductor** is a material whose resistivity becomes zero at and below its critical temperature T_c; **superconductivity** is this phenomenon of zero resistivity below the critical temperature. Uses: strong electromagnets (MRI, particle accelerators), maglev trains and lossless power transmission.

20 **Resistance** is the ratio of the potential difference across a conductor to the current through it, $R = \frac{V}{I}$.

21 **Variation with temperature:** for a metal the resistivity increases with temperature because lattice vibrations increase the collision rate; for a semiconductor the resistivity decreases with temperature because the number of charge carriers increases rapidly.

22 **Stretching a wire:** at constant volume, if the length is multiplied by k, the area is divided by k, so the resistance is multiplied by k^2.

CHAPTER 16 — DIRECT CURRENT CIRCUITS

16.1 Internal Resistance

1 **Electromotive force (e.m.f.)** of a source is the energy converted from other forms into electrical energy per unit charge passing through the source; it equals the terminal p.d. when no current flows.

2 **Internal resistance** of a source is the resistance to the flow of current within the source itself, arising from the materials of the cell.

3 **Potential difference** between two points in a circuit is the electrical energy converted to other forms of energy when a unit charge passes from one point to the other.

4 A **shunt** is a resistor of very low resistance connected in parallel with a galvanometer to divert most of the current, converting it into an ammeter.

5 A **multiplier** is a resistor of very high resistance connected in series with a galvanometer, converting it into a voltmeter.

6 **Terminal potential difference** is the p.d. across the terminals of a source when it delivers current; it is less than the e.m.f. by the 'lost volts' Ir.

7 **Maximum power theorem:** the power delivered to the external resistor is maximum when the external resistance equals the internal resistance of the source, R = r.

8 **Cells in series** have a total e.m.f. and total internal resistance equal to the sums of the individual values (for cells in the same sense).

16.2 Kirchhoff's Laws

9 **Kirchhoff's first law:** the sum of the currents entering a junction equals the sum of the currents leaving the junction. It is based on the conservation of charge.

10 **Kirchhoff's second law:** in any closed loop the algebraic sum of the e.m.f.s equals the algebraic sum of the products IR (potential drops) in the loop. It is based on the conservation of energy.

11 **Sign convention:** choose a direction for each current and for the loop; an e.m.f. is positive if the loop direction passes through the cell from the negative to the positive terminal, and IR is positive if the loop direction is the same as the current direction.

12 **Opposing cells** in a loop give a net e.m.f. equal to the difference of the e.m.f.s.

16.3 Potential Divider

13 **A potential divider** is a circuit in which two or more resistors in series are connected across a supply so that the supply voltage is shared in proportion to the resistances; the output is taken across one of them.

14 **Output voltage** is directly proportional to the resistance across which it is taken, so the larger resistor receives the larger share.

15 **Loading effect:** a load connected in parallel with the output resistor reduces the effective resistance of that arm, so the output p.d. decreases below the no-load value.

16 **Sensors:** the resistance of a thermistor (NTC) decreases as temperature increases, and the resistance of an LDR decreases as light intensity increases; the p.d. across the sensor in a divider therefore decreases.

17 **A rheostat as a potential divider** has its whole track across the supply, which allows the output to vary continuously from zero to the supply p.d.

16.4 Potentiometer and Wheatstone Bridge

18 **A potentiometer** is a device used to measure the e.m.f. of a cell, or to compare e.m.f.s, by balancing the p.d. across a length of uniform wire against the e.m.f. of the cell, so that no current is drawn from the cell at balance.

19 **At the balance point** the galvanometer reads zero, so there is no current in the galvanometer branch and no p.d. across it.

20 **The potentiometer** is more accurate than a voltmeter because at balance no current is drawn from the cell, so the measured value equals the e.m.f. and not the terminal p.d.

21 **A Wheatstone bridge** is balanced when the galvanometer reads zero, which occurs when the ratio of the resistances in one arm equals the ratio in the other, $\frac{P}{Q} = \frac{R}{S}$.

22 **At balance** the two points connected to the galvanometer are at the same potential.

CHAPTER 17 — MAGNETIC FIELDS

17.1 Concept of a Magnetic Field

1 **Magnetic field** is a region in which a moving charge, a current-carrying conductor or a magnetic material experiences a magnetic force.

2 **Magnetic field lines** are lines whose tangent at any point gives the direction of the magnetic field (flux density) at that point; outside a magnet they are directed from the north pole to the south pole.

3 **Magnetic flux density, B,** is the force acting per unit length on a straight conductor carrying a current of 1 A that is placed perpendicular to the magnetic field.

4 **One tesla** is the magnetic flux density that produces a force of 1 N on 1 m of straight conductor carrying a current of 1 A perpendicular to the field.

5 For a moving charge: B is the force per unit charge per unit velocity on a charge moving perpendicular to the field, $B = \frac{F}{qv \sin \theta}$.

6 **Relative permeability** μ_r of a medium is the ratio of the permeability of the medium to the permeability of free space, $\mu_r = \frac{\mu}{\mu_0}$.

7 Magnetic field lines always form closed loops (no magnetic monopoles); they never cross; the closer the lines, the stronger the field.

8 A magnetic field is produced by a moving charge (a current) or by a permanent magnet; a stationary charge produces only an electric field.

9 **Right-hand grip rule (straight wire):** grip the wire with the right hand, thumb pointing along the conventional current; the curled fingers give the direction of the field lines (concentric circles around the wire).

17.2 Force on a Moving Charge

10 **Magnetic force on a moving charge:** a charge q moving with velocity v in a magnetic field B experiences a force of magnitude $F = qvB \sin\theta$, which is perpendicular to both v and B.

11 The magnetic force is always perpendicular to the velocity, so it does no work and does not change the speed or kinetic energy of the particle; it changes only the direction of motion.

12 When v is perpendicular to B, the magnetic force provides the centripetal force and the particle moves in a circle; when v is parallel (or antiparallel) to B, the force is zero and the motion is a straight line; at other angles the path is a helix.

13 **Fleming's left-hand rule:** first finger = direction of B, second finger = direction of conventional current (direction of motion of a positive charge), thumb = direction of force. For a negative charge the second finger points opposite to its velocity.

14 Positive and negative charges of the same velocity are deflected in opposite directions.

15 **Velocity selector:** in perpendicular electric and magnetic fields, a particle passes undeflected when $qE = qvB$, so only particles of speed $v = \frac{E}{B}$ emerge.

17.3 Force on a Current-Carrying Conductor

16 A straight conductor of length L carrying a current I in a magnetic field B experiences a force F = BIL sinθ, perpendicular to both the conductor and the field.

17 **Fleming's left-hand rule:** first finger = field B, second finger = conventional current I, thumb = force F (motion); the three are mutually perpendicular.

18 The force is a maximum (BIL) when the conductor is perpendicular to B and zero when the conductor is parallel to B.

19 The force arises because the field exerts a force on each moving charge carrier; the sum over all the carriers in the length L gives the force on the conductor.

20 A rectangular coil of N turns in a uniform field experiences no net force but a torque τ = NIAB sinφ, where φ is the angle between the normal to the coil and B (equivalently NIAB cosα where α is the angle between the plane of the coil and B).

21 The torque is a maximum when the plane of the coil is parallel to B and zero when the plane is perpendicular to B; this is the principle of the moving-coil meter and the d.c. motor.

17.4 Magnetic Fields Due to Currents

22 **Ampère's law**: for any closed path, the sum of the products of the length elements Δl and the component of B parallel to them equals μ_0 times the current enclosed by the path, $\Sigma B_{\parallel} \Delta l = \mu_0 I$.

23 **Right-hand grip rule (straight wire):** thumb along the conventional current, fingers curl in the direction of B.

24 **Right-hand grip rule (coil/solenoid):** fingers curl in the direction of the current, thumb points along B inside the coil, i.e. towards the north-pole end.

25 The field of a long straight wire consists of concentric circles; B is inversely proportional to the perpendicular distance r.

26 The field inside a long solenoid (far from the ends) is uniform and parallel to the axis; the field outside is very small.

27 Superposition: the resultant field at a point is the vector sum of the fields produced by each current; fields from different wires may cancel at a neutral point.

28 Two coaxial coils or concentric coils with opposing currents give zero net field at the centre when their individual fields are equal in magnitude.

17.5 Force Between Two Current-Carrying Conductors

29 The field of one wire exerts a force on the current in the other; by Newton's third law the two forces are equal in magnitude and opposite in direction, whatever the sizes of the currents.

30 Parallel wires carrying currents in the same direction attract each other; wires carrying currents in opposite directions repel each other.

31 **The ampere** is that constant current which, if maintained in two straight parallel conductors of infinite length and negligible cross-section placed one metre apart in a vacuum, produces a force of 2×10^{-7} N per metre of length between them.

32 Explanation: wire 1 produces a field $B_1 = \frac{\mu_0 I_1}{2\pi d}$ at wire 2; the force on wire 2 is $F = B_1 I_2 L$, directed by Fleming's left-hand rule towards wire 1 when the currents are parallel.

33 **The tesla** is the magnetic flux density that gives a force of 1 N per metre on a conductor carrying 1 A perpendicular to the field.

17.6 Determination of the Ratio e/m

34 **Specific charge** of the electron is the ratio of its charge to its mass, $\frac{e}{m}$ ($C\ kg^{-1}$).

35 Thomson's method: electrons are accelerated and pass through crossed electric and magnetic fields; the fields are adjusted so that the beam is undeflected, so that $eE = evB$, giving $v = \frac{E}{B}$.

36 With the electric field switched off, the magnetic force makes the electrons move in a circular arc: $evB = \frac{mv^2}{r}$, so $\frac{e}{m} = v/(Br)$.

37 Alternative method: electrons accelerated through p.d. V gain kinetic energy $eV = \frac{1}{2}mv^2$; combined with $r = \frac{mv}{eB}$ this gives $\frac{e}{m} = 2V/(B^2r^2)$.

38 The experiment determines the ratio $\frac{e}{m}$ only; the charge e must be found independently (Millikan's oil-drop experiment) to find m.

39 The ratio $\frac{e}{m}$ for cathode rays is the same for all cathode materials, showing that electrons are a universal constituent of matter.

17.7 Hall Effect

40 **Hall effect** is the production of a potential difference (Hall voltage) across a current-carrying conductor or semiconductor placed in a magnetic field, the potential difference being perpendicular to both the current and the field.

41 **Hall voltage** is the transverse potential difference produced across a current-carrying conductor when a magnetic field is applied perpendicular to the current, as the drifting charge carriers are deflected by the magnetic force.

42 Explanation: the magnetic force deflects the moving charge carriers to one side of the slab; the accumulated charge sets up a transverse electric field E_H.

43 The Hall voltage reaches a steady value when the electric force on a carrier balances the magnetic force: $eE_H = evB$.

44 The sign of the Hall voltage identifies the sign of the charge carriers: with the same current direction and field direction, electrons and holes are deflected to the same side but give opposite polarity across the slab (use Fleming's left-hand rule with conventional current).

45 In a semiconductor the carrier density n is much lower than in a metal, so V_H is much larger.

46 Application: a Hall probe measures B because $V_H \propto B$ at constant current; it is made from a thin semiconductor slab for a large V_H.

CHAPTER 18 — ELECTROMAGNETIC INDUCTION

18.1 Magnetic Flux

1 **Magnetic flux** through a surface is the product of the magnetic flux density and the area of the surface, measured perpendicular to the field, $\Phi = BA\cos\theta$; it can be pictured as the total number of magnetic field lines passing through the area.

2 **Weber (Wb)** is the magnetic flux when a magnetic flux density of 1 T passes perpendicularly through an area of 1 m^2.

3 **Magnetic flux linkage** is the product of the number of turns of a coil and the magnetic flux through each turn, $N\Phi$.

4 **Magnetic flux density** B is the magnetic flux per unit area perpendicular to the field (the number of field lines per unit normal area), unit tesla ($T = Wb\ m^{-2}$).

5 For a non-uniform field, the flux is found by summing the normal component of B over all area elements, $\Phi = \int B \cdot dA$.

18.2 Faraday's Law and Lenz's Law

6 **Faraday's law**: the magnitude of the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage (or the rate of cutting of magnetic flux).

7 **Electromagnetic induction** is the production of an induced e.m.f. (and an induced current in a closed circuit) when the magnetic flux linked with a conductor changes.

8 **Induced e.m.f.** is the e.m.f. produced in a conductor when there is a change in the magnetic flux linked with it.

9 **Lenz's law**: the direction of the induced current is such that it opposes the change in magnetic flux that produces it.

10 Lenz's law is a consequence of the conservation of energy: work must be done against the opposing effect to produce the induced current.

11 Induced e.m.f. appears whenever the flux linkage changes, by changing B, A or θ, whether or not the circuit is closed; an induced current flows only in a closed circuit.

12 For a straight conductor moving across a uniform field, the induced e.m.f. is the product of B, the length of the conductor and the velocity perpendicular to B.

18.3 Self Induction

13 **Self-induction** is the production of an induced e.m.f. in a coil due to a change in the current in the same coil, which changes the magnetic flux linked with it.

14 **Self-inductance** L of a coil is the ratio of the magnetic flux linkage in the coil to the current flowing through it, $L = \frac{N\Phi}{I}$.

15 **Henry (H)** is the self-inductance of a coil in which an e.m.f. of 1 V is induced when the current changes at the rate of 1 A s^{-1}.

16 Self-inductance can also be defined as the back e.m.f. induced in a coil per unit rate of change of current in the coil, $L = -\varepsilon / \frac{dI}{dt}$.

17 **Back e.m.f.** is the induced e.m.f. produced in a circuit by a change in its own current; it opposes the change in current that produces it (Lenz's law).

18 By Lenz's law, the self-induced e.m.f. always opposes the change in current: it opposes an increase in current and opposes a decrease in current.

19 The self-induced e.m.f. is large when the current is switched off very quickly, since $\frac{dI}{dt}$ is large; this explains the spark across the switch contacts.

18.4 Energy Stored in an Inductor

20 The energy stored in an inductor is the work done against the self-induced e.m.f. in establishing the current; it is stored in the magnetic field of the inductor.

21 The energy is released when the current decreases, as the self-induced e.m.f. then maintains the current for a short time.

22 The energy stored per unit volume of a magnetic field is the energy density, $u = \frac{B^2}{2\mu_0}$.

23 Energy is stored in the magnetic field of the inductor, not lost; in an ideal inductor there is no energy dissipation.

18.5 Mutual Induction

24 **Mutual induction** is the production of an induced e.m.f. in one coil due to a change in the current in a neighbouring coil, which changes the magnetic flux linked with the first coil.

25 **Mutual inductance** M is the ratio of the magnetic flux linkage in one coil to the current in the other coil that produces the flux, $M = \frac{N_2\Phi_{21}}{I_1}$.

26 **Henry (H)** is the mutual inductance of two coils when a current changing at the rate of 1 A s^{-1} in one coil induces an e.m.f. of 1 V in the other.

27 Mutual inductance can also be defined as the e.m.f. induced in one coil per unit rate of change of current in the other coil, $M = -\varepsilon_2 / \frac{dI_1}{dt}$.

28 Mutual inductance depends on the geometry of the coils (number of turns, size, separation, relative orientation) and the permeability of the core, not on the current.

29 **Transformer**: an alternating current in the primary coil produces a changing flux in the core, which induces an alternating e.m.f. in the secondary coil by mutual induction.

CHAPTER 19 — ALTERNATING CURRENT CIRCUITS

19.1 AC Through a Pure Resistor

1 An **alternating current** is a current that reverses its direction at a constant frequency; usually it varies sinusoidally with time, $I = I_0 \sin\omega t$.

2 **Root-mean-square (r.m.s.) value** of an alternating current is the value of the steady direct current that dissipates heat in a given resistor at the same rate as the alternating current.

3 In a pure resistor, the current is in phase with the voltage.

4 The instantaneous power $P = I^2R$ varies with a frequency twice that of the current and is always non-negative; the mean power is half the peak power.

5 Peak value: the maximum magnitude of the current or voltage during a cycle.

19.2 AC Through a Pure Inductor

6 **Inductive reactance** X_L is the opposition of an inductor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the inductor to the r.m.s. (or peak) current through it.

7 In a pure inductor, the current lags the applied voltage by 90° ($\frac{\pi}{2}$ rad).

8 The mean power dissipated in a pure inductor is zero over one complete cycle, because energy is stored in the magnetic field and returned to the source alternately.

9 At very high frequency, the inductor opposes the current strongly; for direct current ($f = 0$), $X_L = 0$ and the inductor offers no opposition (apart from its resistance).

19.3 AC Through a Pure Capacitor

10 **Capacitive reactance** X_C is the opposition of a capacitor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the capacitor to the r.m.s. (or peak) current through it.

11 In a pure capacitor, the current leads the applied voltage by 90° ($\frac{\pi}{2}$ rad).

12 The mean power dissipated in a pure capacitor is zero over one complete cycle, because energy is stored in the electric field and returned to the source alternately.

13 A capacitor blocks direct current ($X_C \to \infty$ as $f \to 0$) and passes high-frequency alternating current more easily ($X_C \to 0$ as $f \to \infty$).

19.4 R-C and R-L Series Circuits

14 **Impedance** Z is the total opposition of an a.c. circuit to the alternating current; it is the ratio of the r.m.s. (or peak) voltage across the circuit to the r.m.s. (or peak) current in it.

15 In a series R-L circuit, the voltage leads the current by a phase angle φ; in a series R-C circuit, the current leads the voltage by φ, where $0° < \varphi < 90°$.

16 **Power factor** is the ratio of the mean power dissipated to the product of V_{rms} and I_{rms}, equal to $\cos\varphi = \frac{R}{Z}$.

17 The mean power is dissipated only in the resistance; the inductor and capacitor dissipate no mean power.

18 The voltages across R and L (or C) are 90° out of phase, so the supply voltage is their phasor (vector) sum and not their arithmetic sum.